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localhost:8889/notebooks/HillClimbing_N_Queens.ipynb

jupyter HillClimbing_N_Queens Last Checkpoint: 20 minutes ago

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Code

JupyterLab Python 3 (ipykernel)

```
    while True:
        cost = calculate_cost(board)
        print("State:", board, "Cost:", cost)

        if cost == 0:
            print("Goal Reached!")
            return board

        best_board = board[:]
        best_cost = cost

        for col in range(n):
            for row in range(n):
                if row != board[col]:
                    neighbor = board[:]
                    neighbor[col] = row
                    new_cost = calculate_cost(neighbor)

                    if new_cost < best_cost:
                        best_cost = new_cost
                        best_board = neighbor[:]

        if best_cost >= cost:
            print("Local Minimum!")
            return None

        board = best_board


# Main
initial_board = [0, 0, 0, 0]
solution = hill_climbing(initial_board)

print("Final Solution:", solution)
```

```
State: [0, 0, 0, 0] Cost: 6
State: [0, 3, 0, 0] Cost: 3
State: [1, 3, 0, 0] Cost: 1
State: [1, 3, 0, 2] Cost: 0
Goal Reached!
Final Solution: [1, 3, 0, 2]
```

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